

Analytics Diagnostic Report: OEE

SECTION 1

Executive Summary

Plant Overall Equipment Effectiveness (OEE) stands at **64.0%**, against a target of 85%. The shortfall is split roughly evenly between **Availability** (81.9%), the time lost to unplanned stoppages, alarms, and changeovers, and **Performance** (82.2%), machines running below their ideal cycle speed. The Quality rate of 95% is an assumption due to a lack of proper data collection on quality outcomes, so it is not measured directly here.

64%

PLANT OEE

82%

AVAILABILITY

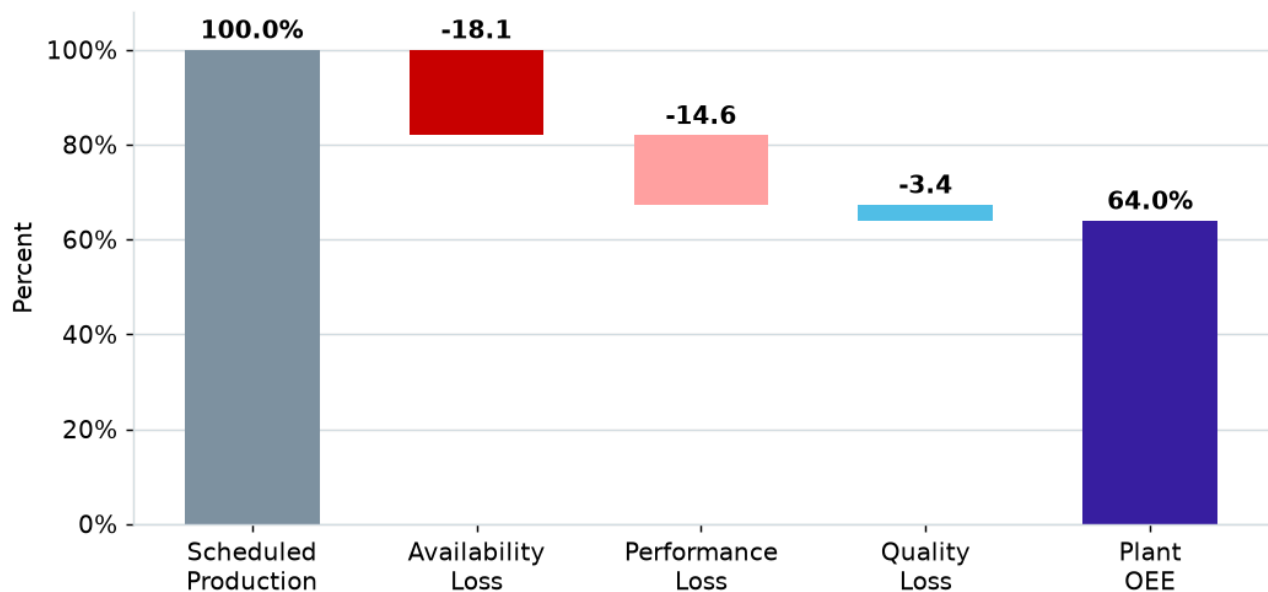
82%

PERFORMANCE

95%

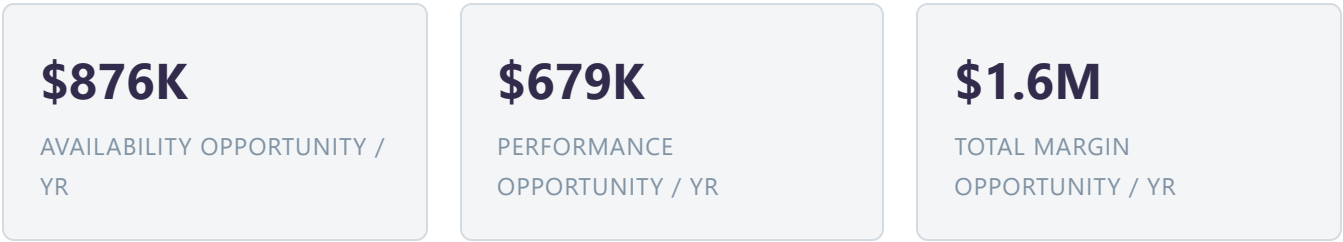
QUALITY (EST.)

How Plant OEE Breaks Down



Reaching the 85% target OEE (implying availability of **94%** and performance of **95%**) is worth an estimated **\$1.6M per year** in recovered contribution margin. The opportunity divides into an availability opportunity of **\$876K** (less downtime and faster changeovers) and a performance opportunity of **\$679K** (running nearer ideal speed). Within availability, unplanned

reactive repairs alone account for **\$325K per year** and are the most directly addressable option available. Section 5 details the breakdown of this contribution margin uplift opportunity.

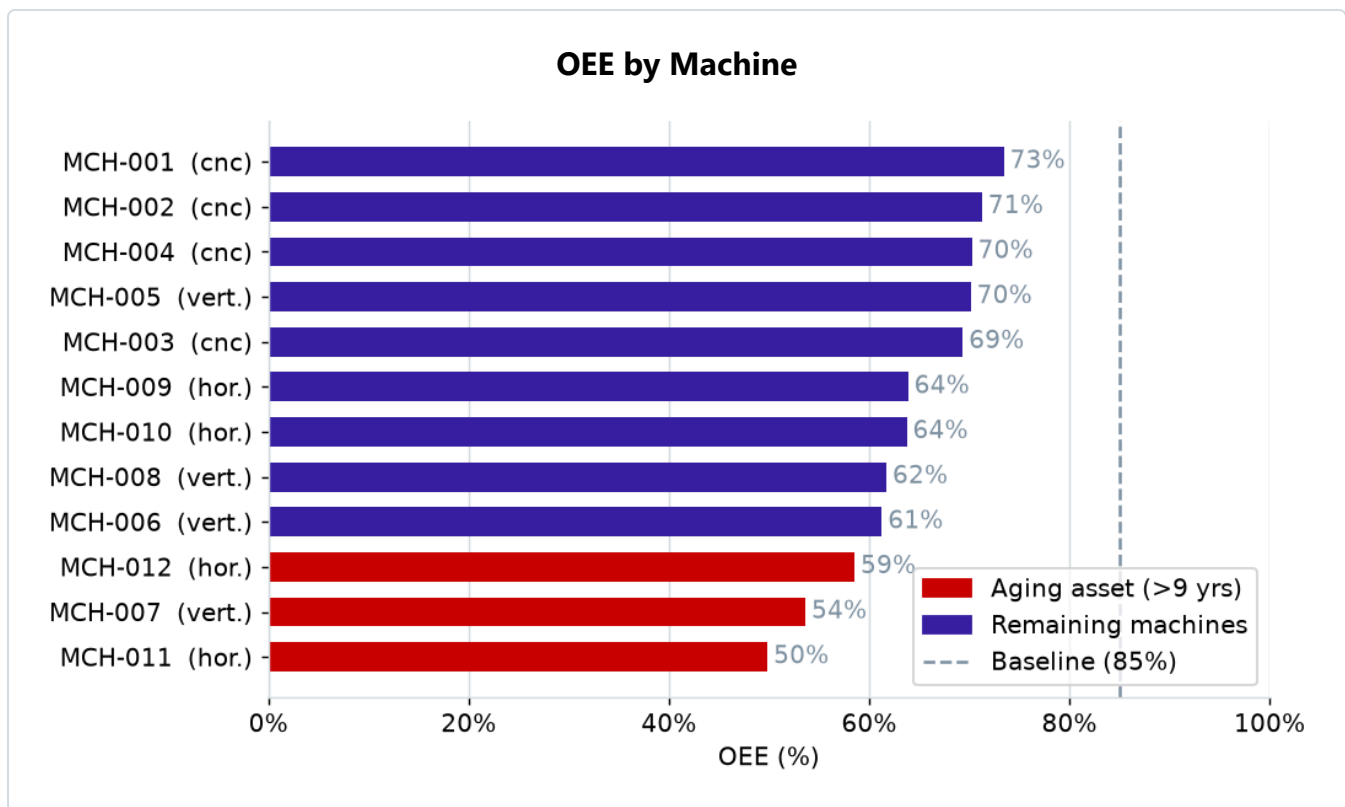


Three findings account for most of the gap between today's 64.0% OEE and the 85% target. First, the two aging horizontal mills, **MCH-011, MCH-007**, lose OEE on both levers at once: they run slower and stop more often than the rest of the shop, so they sit last on every ranking. Second, tooling is the single largest source of unplanned downtime, accounting for 45% of all unplanned downtime hours and \$146K a year in lost contribution margin. Third, unplanned stoppages run about 1.5x higher in the first 45 minutes of a shift than through the rest of it, on both crews, which points at shift-start routines rather than at any one team. The Recommended Actions section closes the report by turning these into a plan, ranked by annual impact.

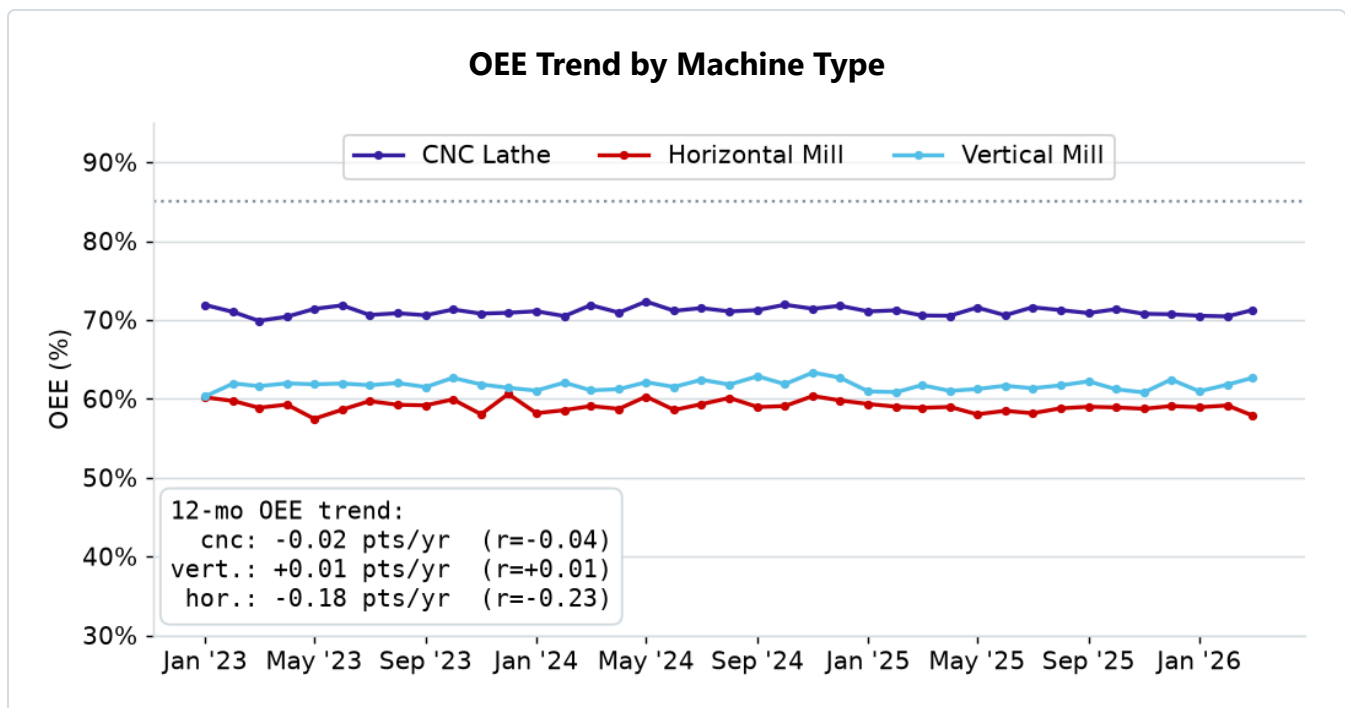
SECTION 2

Machine-level OEE Breakdown

OEE ranges from 73% on the best machine to 50% on the worst. CNC lathes sit at the top of the shop and horizontal mills at the bottom, with the two aging assets, **MCH-011, MCH-007**, ranked last. The pattern is structural: the horizontal mills are the oldest, most capital-intensive machines and take the most complex work, so they lose more time to stoppages and run slower, while the lathes are newer and run simpler jobs.

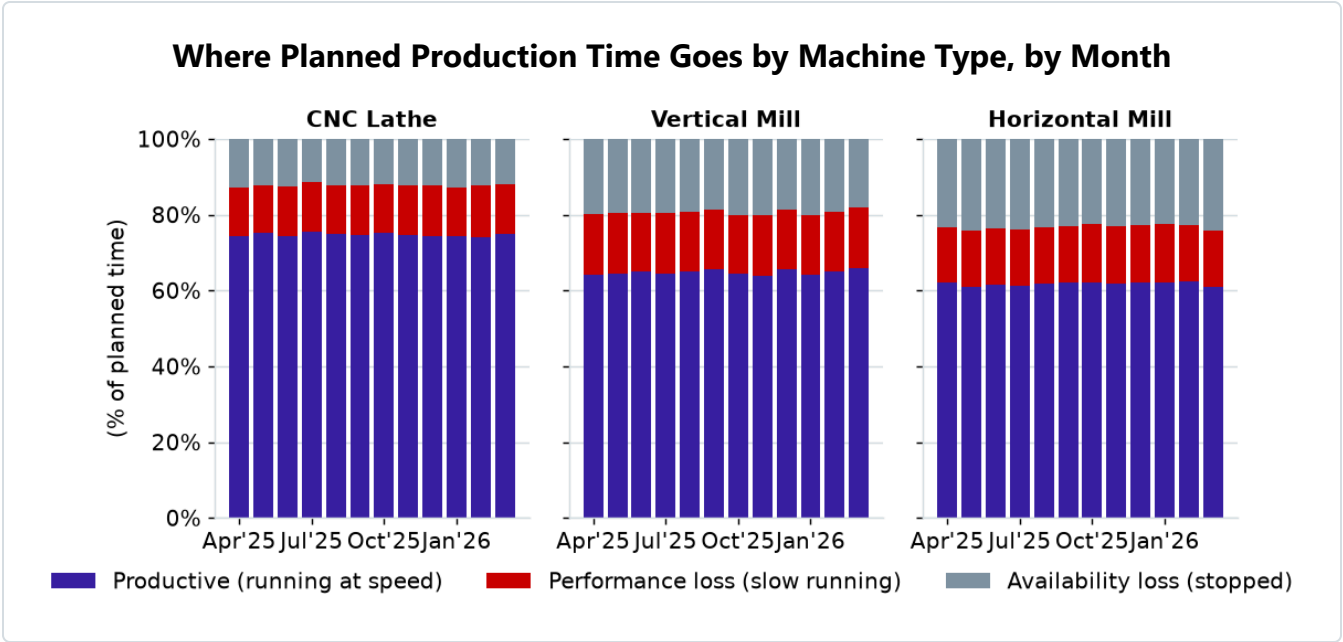


The OEE differences by machine type holds across the whole period rather than reflecting a good or bad month. Fitting a linear trend to each machine type's monthly OEE confirms that the series are statistically flat, with no noticeable increase or decrease over time.



The charts below shows each machine type's planned production time split into three parts: time spent productive (running at target speed), time lost to slow running (a Performance loss),

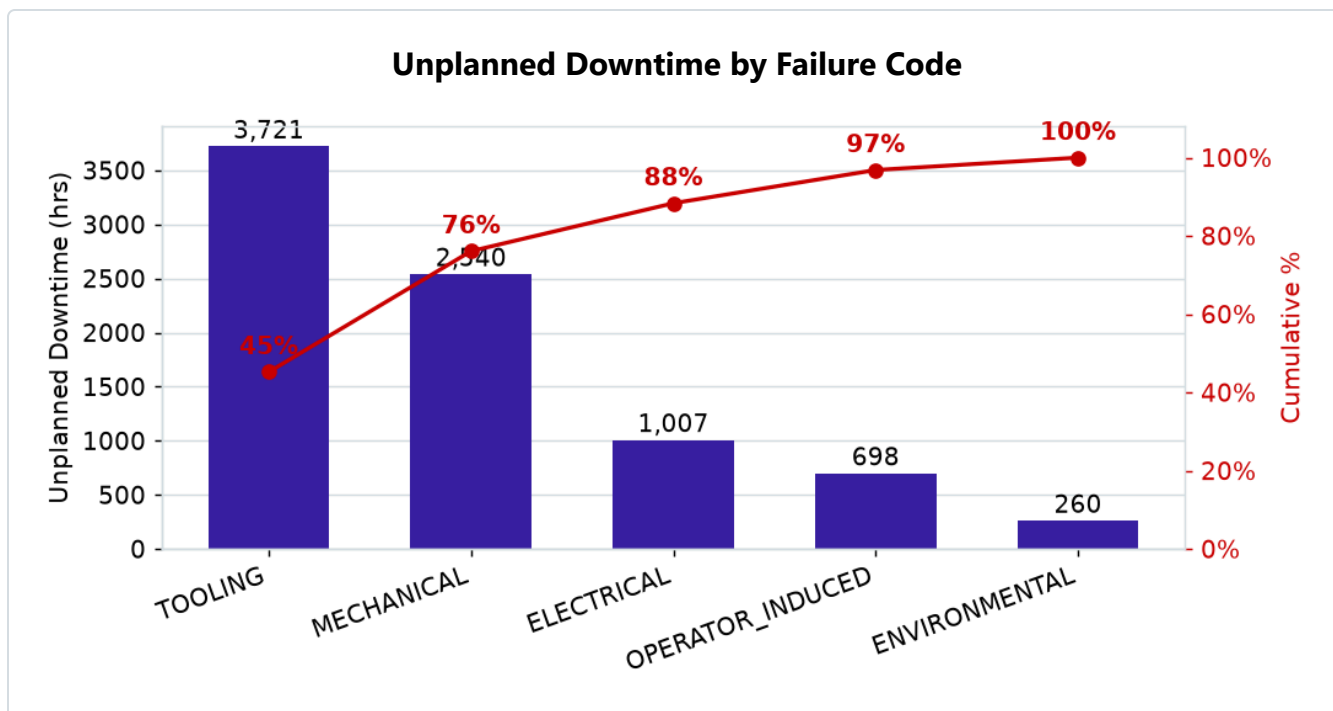
and time lost to stoppages (an Availability loss). The productive share erodes across all machine types, and the erosion splits fairly evenly across both Availability and Performance, so neither lever can be ignored. The production losses hold steady month to month across the past year, which marks these losses as structural rather than seasonal and sets up the two deep dives that follow.



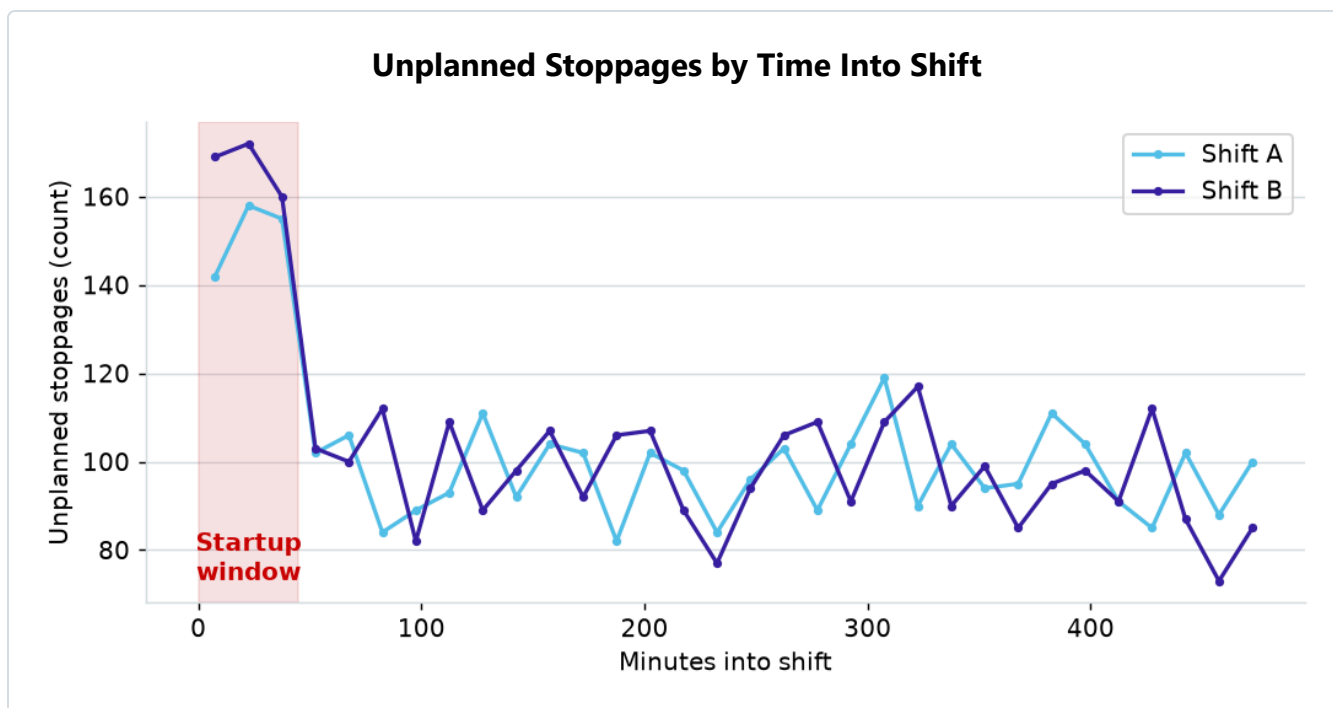
SECTION 3

Deep Dive: Availability

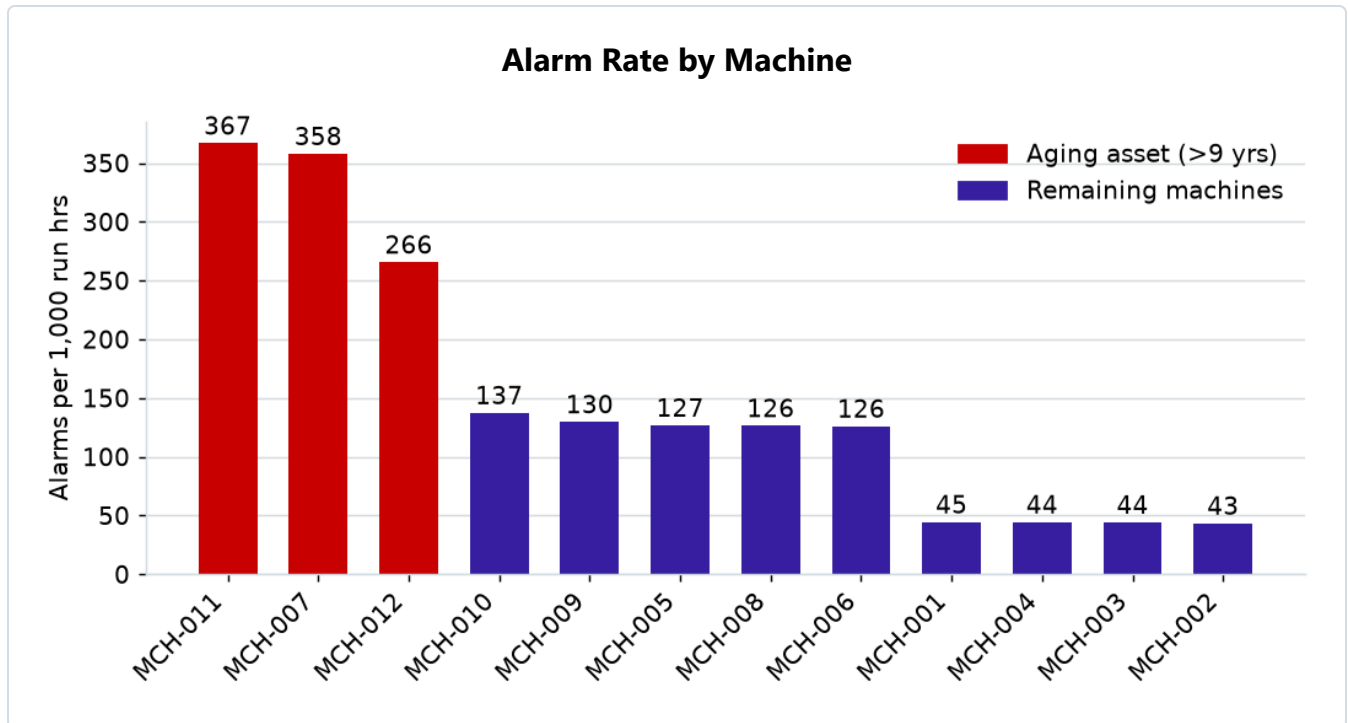
The below downtime analysis shows where Availability is lost and why. Unplanned downtime concentrates heavily in a few failure modes: about 76% of all downtime hours trace to just two, tooling and mechanical. The orange line is the cumulative share of unplanned downtime by failure mode, so a focused program on those first two categories addresses most of the loss.



Downtime is also not spread evenly through the day. Counting when unplanned stoppages begin across the full observation window (Jan 2023-Mar 2026), both crews run steadily through the middle of their shift and both stumble at the start of it: stoppages in the opening 45 minutes run about 1.5x the mid-shift rate, at 1.5x on Shift A and 1.6x on Shift B. That the two shifts look so alike is the point. Cold machines, warm-ups, and first-piece verification, not one weaker team, are what drive the cluster, so the target is the shift-start routine itself: a standardised warm-up and first-piece check written into the opening of every shift.

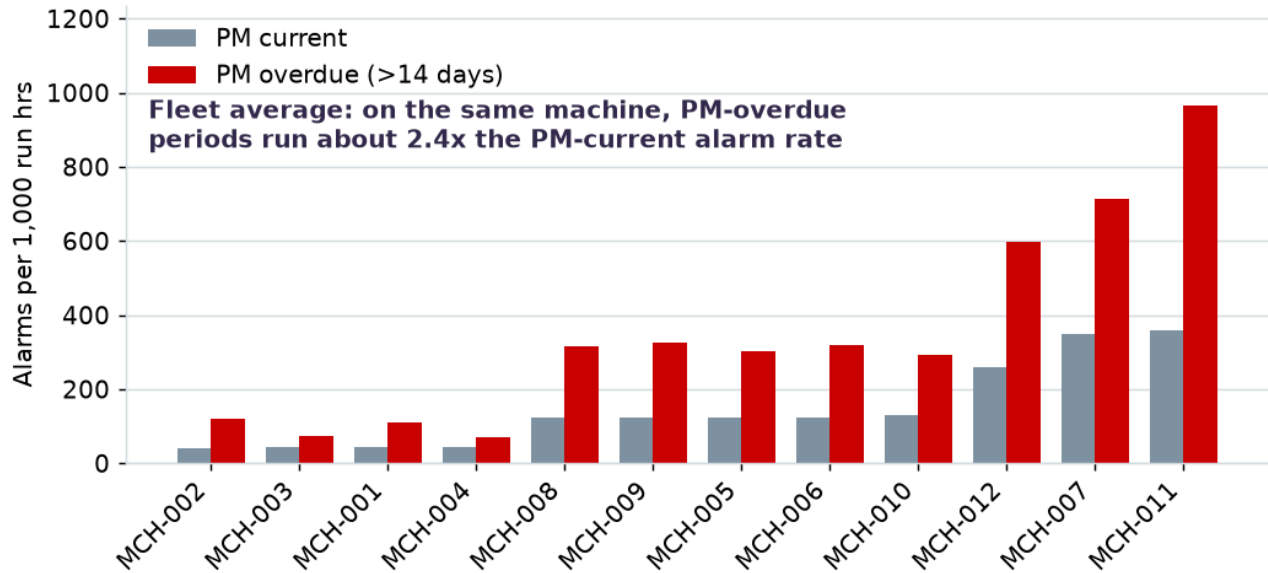


Alarms are the shop's early-warning signal. A machine raises one when it detects a condition it cannot resolve on its own, so alarms cluster in the run-up to a stoppage and a rising alarm rate marks an asset drifting toward failure well before it goes down. Normalised per 1,000 running hours to keep busy and idle machines comparable, the burden is far from evenly spread: the aging assets raise alarms several times more often than the rest of the shop, which is what puts them first in line for attention.



Reading that alarm rate against each machine's preventive maintenance (PM) history shows how much of the burden is self-inflicted. Comparing an asset against itself is what makes the comparison trustworthy: the same machine, same age, same duty, measured in the spans when it is more than 14 days past a due PM against the spans when its PM is current. On that basis alarms run about **2.4x** higher while a machine is overdue, and the gap holds for newer and older machines alike. Shop-wide on-time PM completion is 72% and uneven, with **MCH-007** at only 62%. Catching up on PM is therefore a reliability lever, not just a compliance metric.

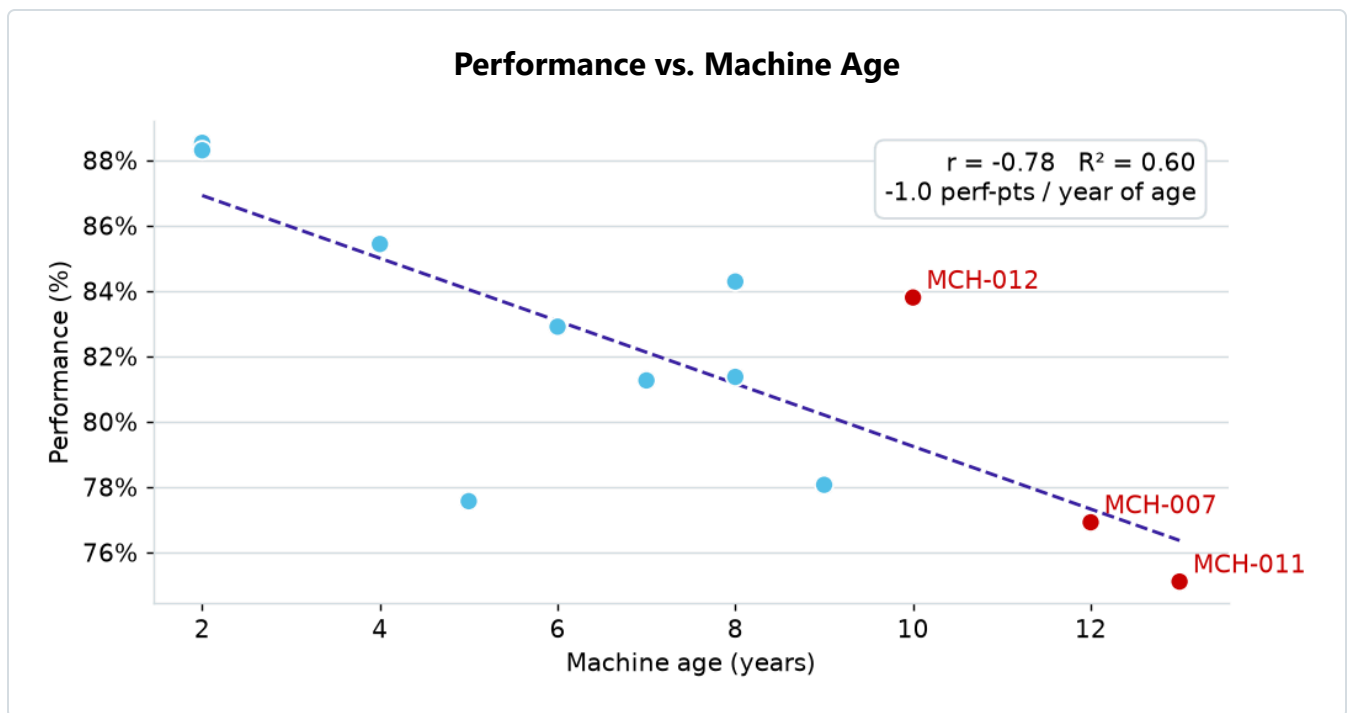
Alarm Rate: PM-Overdue vs. PM-Current Periods, by Machine



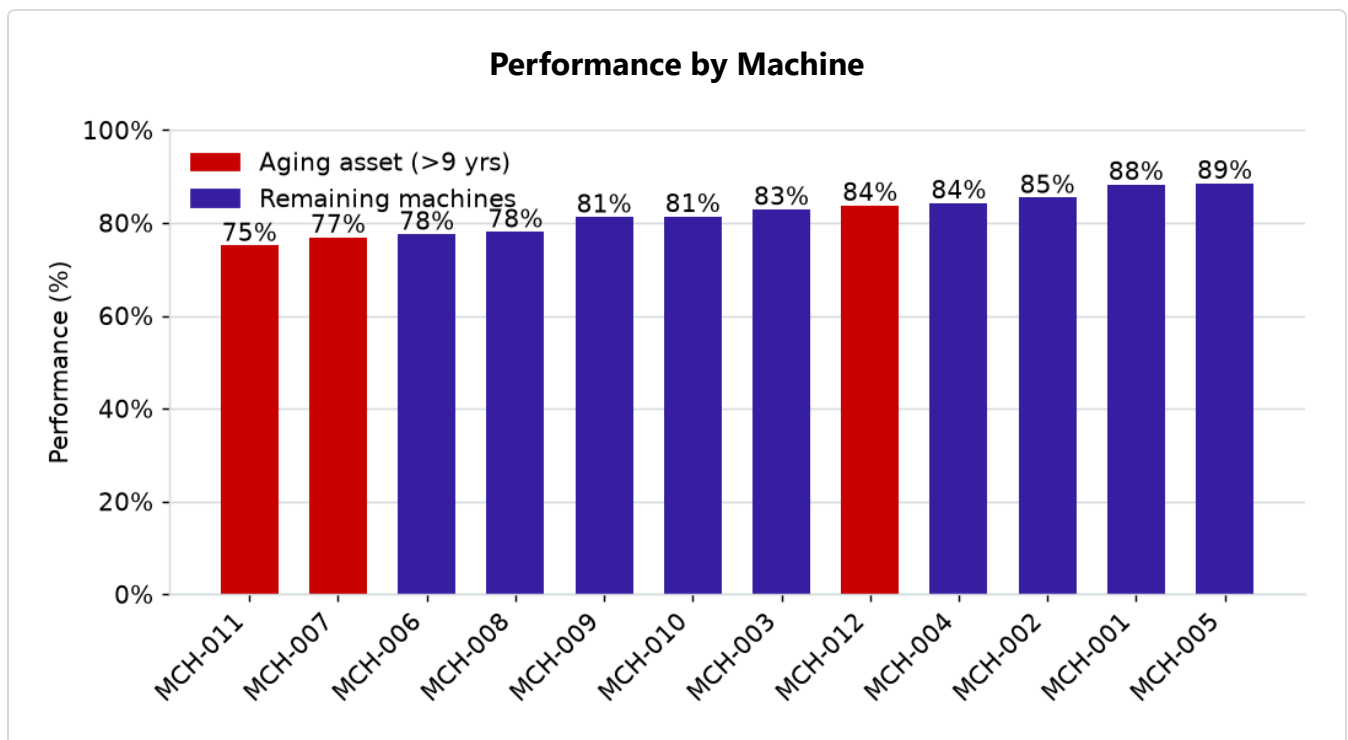
SECTION 4

Deep Dive: Performance

Performance measures actual cycle time against the ideal cycle time a machine is capable of, and here it is closely tied to machine age. Older machines hold a slower cycle through worn spindles, dated controls, and more conservative feeds and speeds. The relationship is strong and statistically clear: across the shop, Performance falls about 1.0 points for every year of machine age. This is why Performance, not just downtime, deserves its own view.

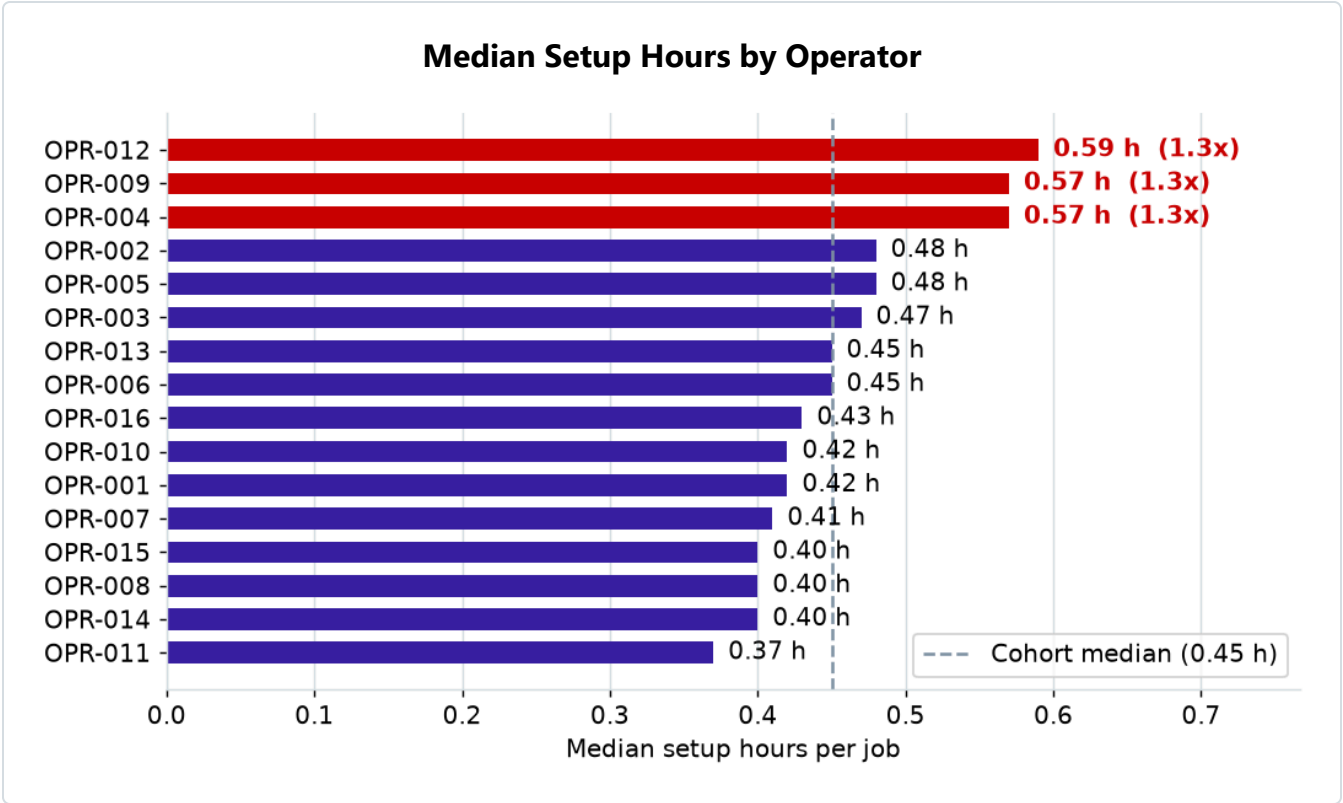


Ranked by machine, the spread is wide: the newest lathes run in the high 80s while the oldest horizontal mills sit in the mid 70s. Because Performance compounds with the lower Availability on those same aging assets, the oldest machines lose OEE on both fronts at once, reinforcing where recapitalisation or spindle rebuilds would pay back fastest.



Setup time is the one Performance lever the shop controls without touching the machines. Every minute spent setting up is a minute the spindle is not cutting. Most machinists cluster

near the cohort median of 27 minutes per job, and the spread across the floor is real but modest. Three of them, **OPR-012, OPR-004, OPR-009**, sit consistently above the rest at about 1.3x the median, roughly 8 minutes longer on every job they run. That is a targeted coaching and setup-standardisation opportunity rather than a machine problem, and it recovers Performance at no capital cost.



SECTION 5

Margin Uplift Opportunity Detail

The OEE gaps translate directly into lost contribution margin, the incremental profit on each productive machine hour (revenue net of variable cost), not top-line revenue. Valued at each machine type's contribution margin (CNC Lathes \$95/hr, Vertical Mills \$125/hr, Horizontal Mills \$145/hr), reaching the 85% target is worth an estimated **\$1,554,807 per year**, and it splits cleanly between the two controllable levers: **\$876,252 per year** from Availability (less downtime and faster changeovers) and **\$678,555 per year** from Performance (running nearer ideal speed). Quality is held at the assumed 95% and is not treated as a lever here. Both Availability and Performance are opportunity costs, realisable to the extent recovered capacity can be sold.

Within the Availability opportunity, unplanned reactive repairs logged in the CMMS are the most visible and directly addressable opportunity, running about **\$324,706 per year** (2,531 hours) on their own; the balance is setup, changeover, and idle time. Within the Performance opportunity, the loss concentrates on the oldest machines, where the operator setup-time gap from Section 4 is a lever that costs nothing to pull.

\$876K

AVAILABILITY OPPORTUNITY /
YR

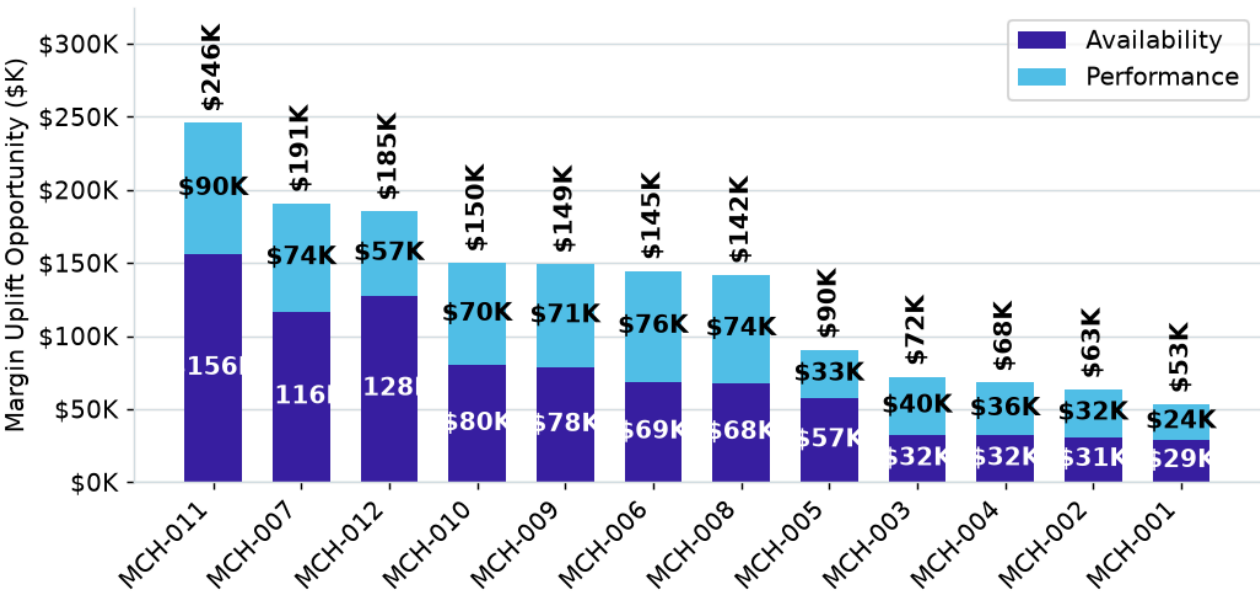
\$679K

PERFORMANCE
OPPORTUNITY / YR

\$1.6M

TOTAL MARGIN
OPPORTUNITY / YR

Contribution margin uplift opportunity at 85% OEE, by Machine



SECTION 6

Recommended Actions

The findings resolve into five actions, ranked below by the annual impact each one carries. The estimates are drawn from the same data and contribution-margin assumptions used throughout this report.

ACTION	SUPPORTING FINDING	EST. ANNUAL IMPACT	CAPITAL REQUIRED
Reliability review, rebuild versus replace, for MCH-011, MCH-007	Aging assets losing on both OEE levers (P1)	\$437K	Capex decision
Tooling program: vendor consolidation and tool-life tracking	Tooling is the largest downtime category, 45% of unplanned hours	\$58K	Low
Standardised shift-start warm-up and first-piece routine on both shifts	Stoppages run 1.5x higher in a shift's first 45 minutes (P2)	\$24K	None
Setup standardisation coaching for OPR-012, OPR-004, OPR-009	Setups at 1.3x the cohort median (P4)	\$10K	None
PM catch-up and compliance enforcement	Within-machine PM-overdue to alarm-rate linkage (P3)	\$8K	Minimal

Impact and effort do not run in the same order. The reliability review sits at the top of the table but is the only item carrying a capital decision, so it needs a business case before it moves. The actions below it require little or no capital and can begin immediately, which makes them the natural first wave while the larger commitment is worked through.